

Jeremy Swanson

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EDUCATION

University of Central Florida

Bachelor of Science in Computer Engineering | GPA: 3.2

Orlando, FL

May 2027

Relevant Courses: Data Structures and Algorithms, Linear Circuits, Electronics, Computer Organization, Computer Architecture, Digital Systems, Discrete Structures, Embedded Systems, Junior Design

PROJECTS

Ultrasonic Range Finder – Individual Project

May 2026 - July 2026

- Designed and fabricated a battery-powered ultrasonic range finder around an MSP430G2553, integrating an HC-SR04 sensor, I2C 16x2 character LCD, PWM-dimmed LED, and potentiometer-driven ADC input on a custom two-layer PCB.
- Created schematics, footprints, and board layouts in Autodesk Fusion 360 - imported vendor libraries via Ultra Librarian, hand-routed traces with ground-plane pours, cleared DRC/ERC, and generated Gerber and pick-and-place files for JLCPCB fabrication.
- Developed embedded C firmware using register-level peripheral configuration: 10-bit ADC sampling to drive PWM duty cycle, timer-based echo pulse-width capture converted to distance in mm and inches, and I2C driver calls to render live measurements to the LCD on a 500ms refresh loop.
- Designed and validated two boost converter PCBs (TI TPS613221/TPS613222) stepping two AA cells up to regulated 3.3V and 5.0V rails at 150mA. Verified efficiency, ripple, and thermal performance against TI WEBench predictions using electronic load, oscilloscope, and thermal camera.

Black Friday Brawlers – Group Project

Aug 2025 - Dec 2025

- Built a 2.5D beat-em-up in Unity set during an early-2000s mall Black Friday, working on a team of 5 across a full semester development cycle.
- Designed and implemented the item system, including item behavior, combat-item spawn logic, and spawn rate balancing across enemy encounter types.
- Developed the in-game HUD, surfacing player health, active items, and combat state through Unity's UI system.

Syllab.AI – Group Project

Apr 2025

- Developed the mobile client for an AI-powered study platform that parses uploaded PDF syllabi into chapter summaries, interactive quizzes, and study materials.
- Built the cross-platform mobile front end in Flutter and Dart.
- Delivered as part of a 6-person team over an 8-week build cycle, deploying to iOS via XCode.

ContactMe – Group Project

Feb 2025

- Developed the front end for a LAMP-stack contact management application supporting user authentication and full CRUD operations on stored contacts.
- Implemented the client interface in HTML, CSS, and JavaScript, wiring UI actions to backend PHP endpoints via asynchronous requests.

WORK EXPERIENCE

Server

Fishlips Waterfront Bar and Grill

Cape Canaveral, FL

Feb 2021 – Present

- Coordinated with front-and back-of-house staff to serve high-volume shifts at a waterfront restaurant, maintaining service quality under sustained peak demand.
- Sustained five years of continuous employment while carrying a full engineering course load.

TECHNICAL SKILLS

Languages: C, C#, Java, Python, JavaScript, Verilog, MATLAB, Dart, HTML, CSS, Assembly (ARM, MIPS)

Embedded Systems: MSP430 (G2553, FR6989), Register-level peripheral programming, ADC, UART, SPI, I2C, GPIO

Frameworks & Libraries: NumPy, Pandas, Flutter, Pygame, ModernGL, OpenGL, Unity, Matplotlib

Computer Architecture: Pipelining, Caching, Memory Hierarchy

Hardware: Oscilloscope, Function Generator, DC Supply, Thermal Imaging Camera, PCB design and layout, SMT reflow assembly, Through-hole soldering and rework

Tools: Git, Figma, Visual Studio Code, Code Composer Studio, XCode, MySQL, Jupyter Notebook, LaTeX, Xilinx Vivado, LTspice, MARS MIPS, Autodesk Fusion 360, Ultra Librarian, TI WEBench